

GrayTalian Reclist Guide

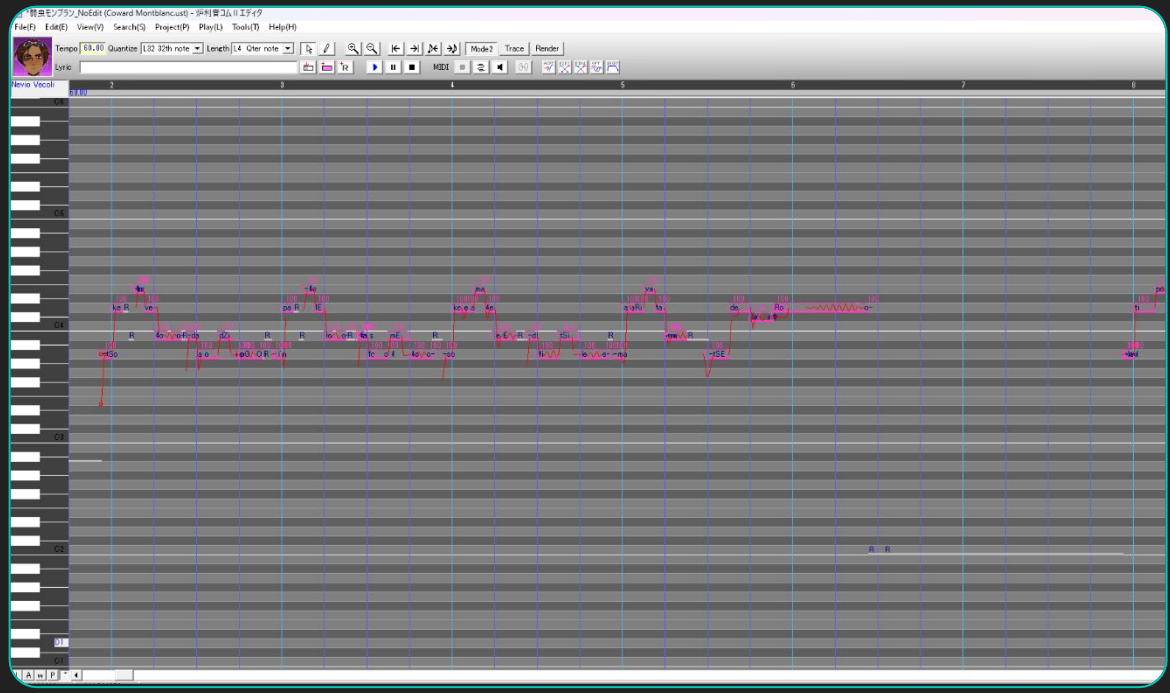
How to use GrayTalian · GraySlate · 2024

Table of Contents

Introduction	OREMO Recording	Setting up OREMO	Recording	X-Sampa Phonemes	Post-Processing
Generating FRQs	Creating OTOs	Creating OTOs (cont'd)	UTAU OTO	UTAU OTO (cont'd)	Standalone Vowels (1/3)
Standalone Vowels (2/3)	Standalone Vowels (3/3)	Vowel-Vowels (VVs)	Main CVs	Hard Consonants (CV)	Soft Consonants (CV)
"S-Heavy" Consonants (CV)	Vowel-Consonants (VCs)	Consonant Clusters	Aliases	Bank Testing	Fitting a UST
Playback					

Introduction

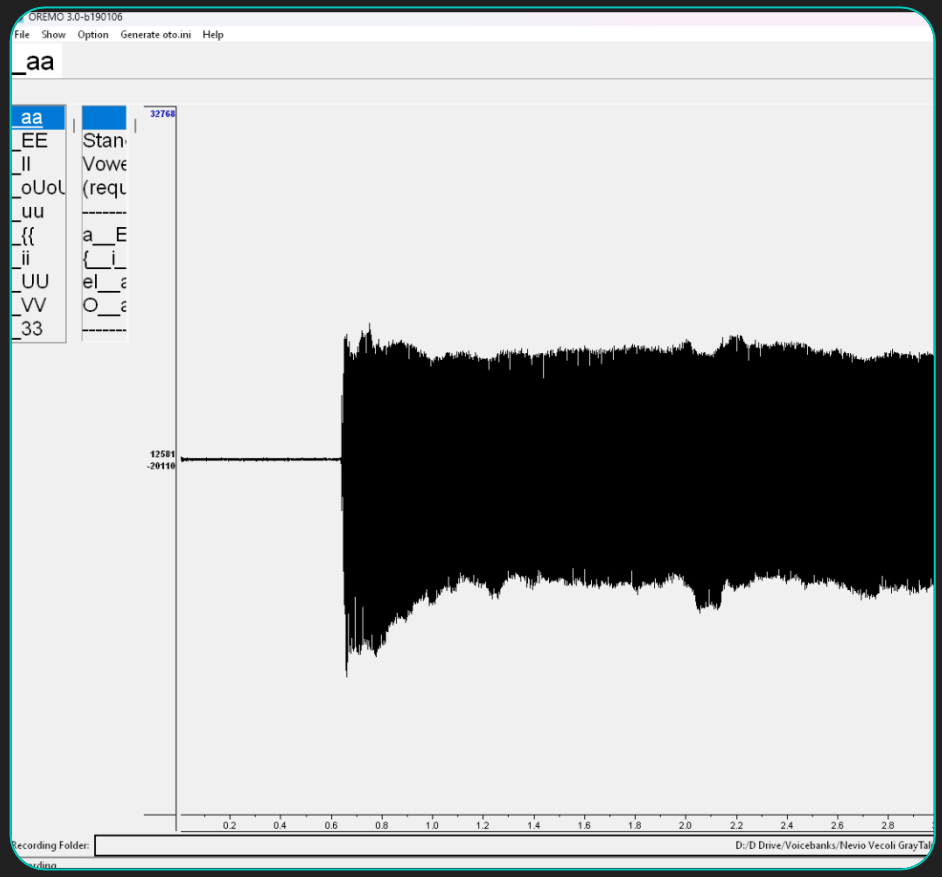
- This guide is intended to guide UTAU users through the process of creating a voicebank on the GrayTalian CVVC Italian reclist. Every essential step will be covered in this guide.

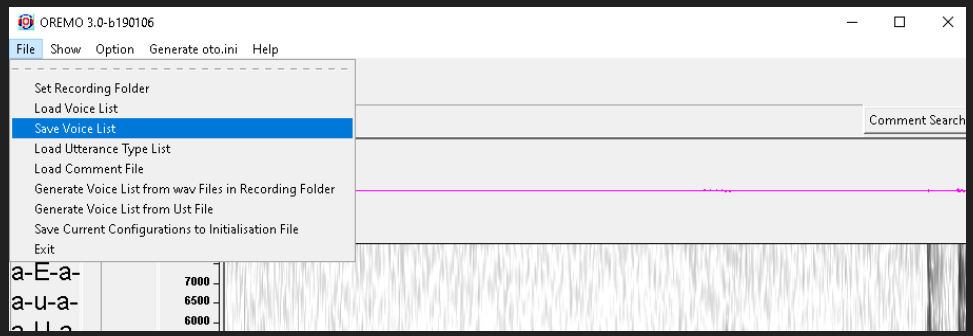


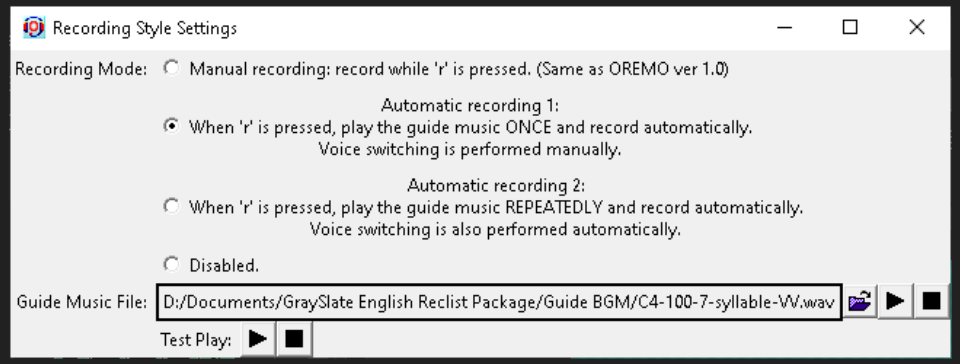
GrayTalian will be the informal name used throughout the rest of this guide.

OREMO Recording

- OREMO is the recommended software for recording a GrayTalian voicebank, as well as UTAU voicebanks in general.
- OREMO in English can be downloaded here
- Several files in the reclist package folder will be needed for recording in OREMO:
  - OREMO reclist.txt
  - X-Sampa phoneme pronunciation guide.PDF
- Guide BGM is optional but any 2-syllable guide BGM will work





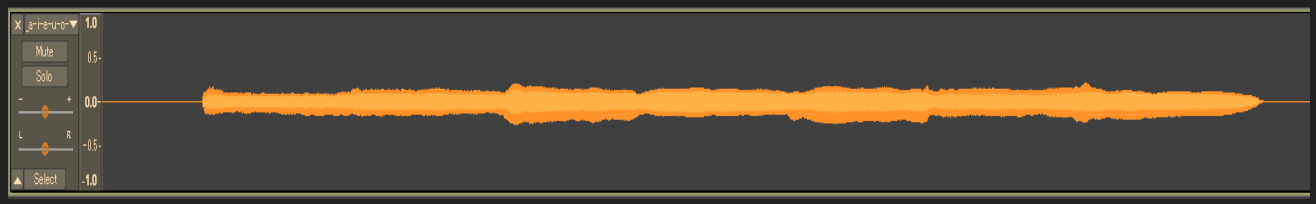


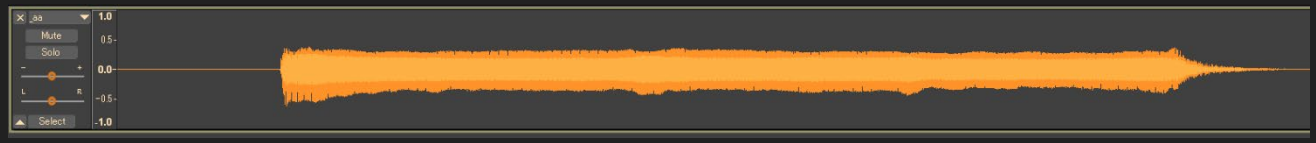
Setting up OREMO

- Set the target recording folder as your voicebank folder by navigating to File > Set Recording Folder and selecting the correct folder.
- Load the reclist into OREMO by navigating to File > Load Voice List and selecting OREMO reclist.txt
- To use a guide BGM with OREMO, navigate to Option > Recording Style Settings. From here, select Automatic Recording 1. Then, click the folder icon next to Guide Music File and navigate to the Guide BGM folder. Select the BGM you would like to use. It is recommended to listen to the guide BGM to get a feel for how the recordings are done.

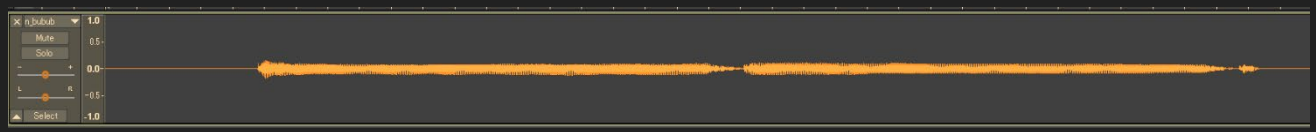
Recording

- The bulk of GrayTalian is 2-syllable strings, with the exception being the beginning –CCs and VV strings.
- Vowel-only strings are recorded as VCV and are continuous.
- Standalone vowel strings are recorded with end breaths.
- -CC and CC/CC- strings can be recorded quickly, i.e. it is not necessary to hold the note for very long.









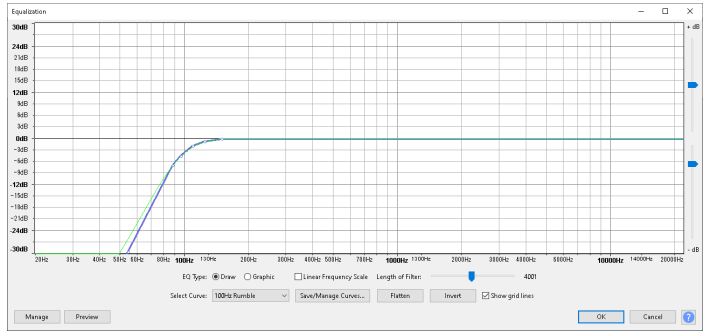
X-Sampa Phonemes

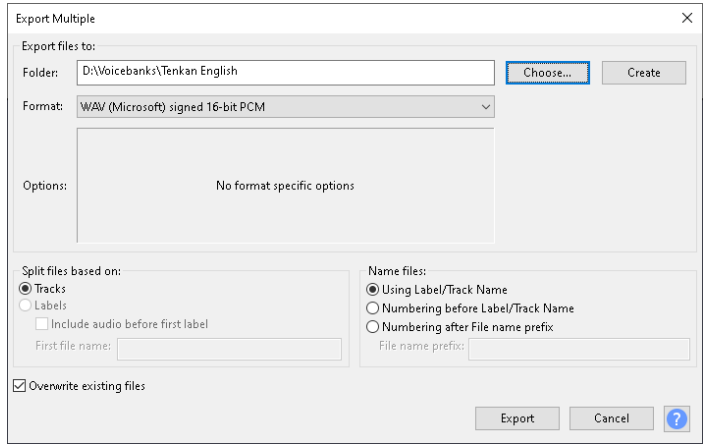
- X-Sampa is the phoneme transcription method that is used in this reclist.
- A phoneme table for X-Sampa phonemes has been provided. This compares phonemes from other reclists, as well as providing example words.
- If at any point you become stuck, I have included a link to a sample bank.

Phoneme Chart		
IPA	**GrayTalian X-Sampa**	**Usage Example**
Vowels		
a	a	C**a**drò
i	i	F**i**nito
u	u	S**u**cesso
ɔ	O	Am**ò**
o	o	Ner**o**
e	e	Perch**é**
ɛ	E	Cio**è**
Consonants		
b	b	S**b**agliare
tʃ	tS	**C**iò
d	d	**D**onna

Post-Processing

- After all recordings are finished, little to no post-processing is recommended.
- To clean up recordings, open the voicebank folder and move only the WAV files to a blank Audacity window.
- Select all files.
- From the Effects tab, select Equalization. In the Select Curve dropdown, choose 100Hz rumble. Click OK.
- After the effect is applied, select File > Export > Export Multiple. Choose your voicebank folder, make sure the format is 16-bit WAV, make sure split files based on tracks is chosen, make sure name files using label/track name is selected, and check the box next to "overwrite existing files." Click export and select OK for all files.





Generating FRQs

Generating FRQs is recommended before setting up the base oto.

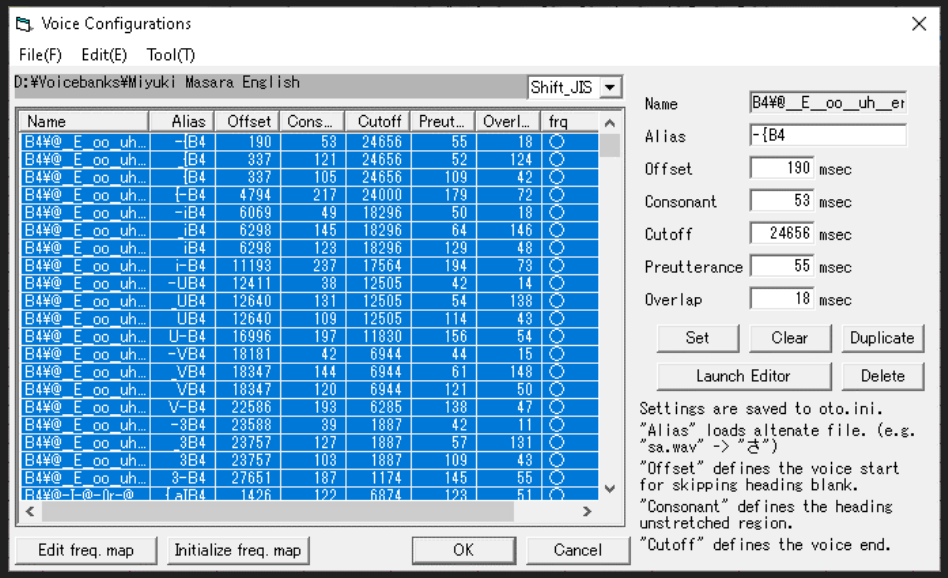
Open UTAU and with your voicebank loaded into UTAU, select Tools > Voicebank Settings. Right click and select "Select Multi," then select all.

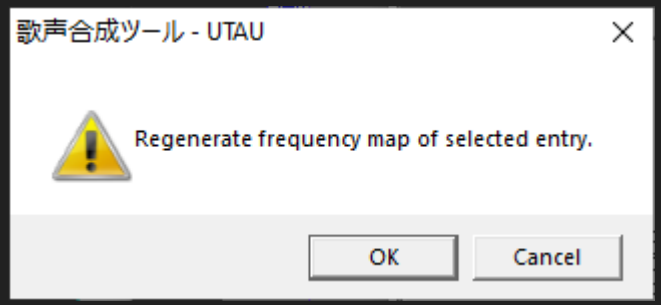
Click "Initialize Frequency map" and OK to generate FRQs for all recordings.

Once done, hit OK to save and close the editor.

TIP: Using the resampler patcher will assist in speeding up FRQ generation.

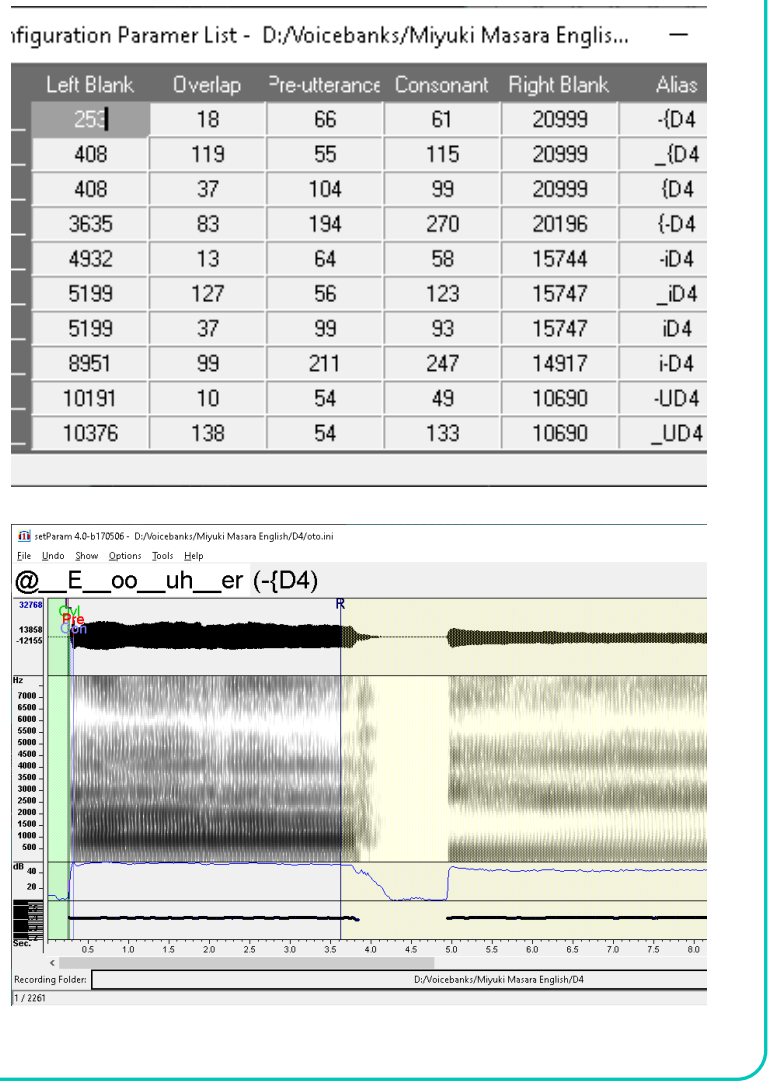
- You can also use nmsao1's frq editor plugin to generate MRQs and convert to FRQs.

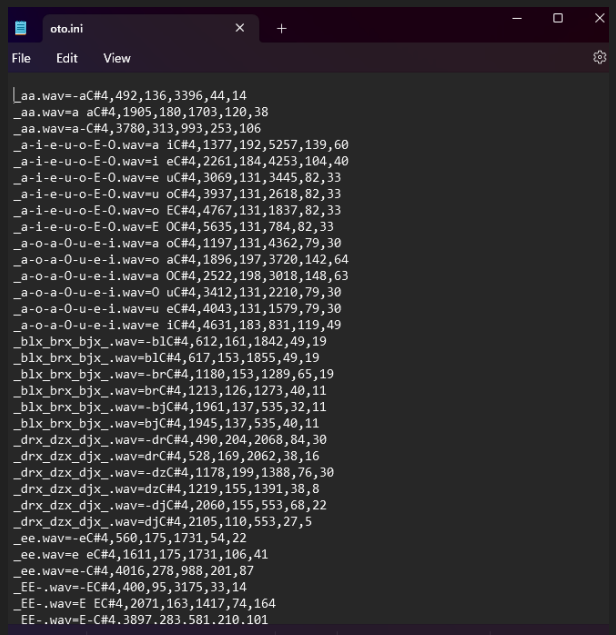


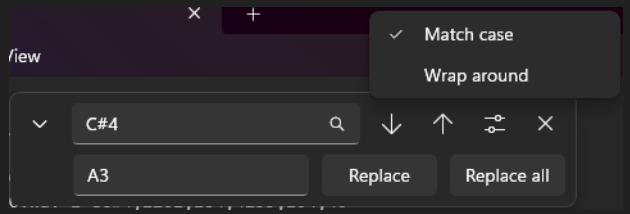


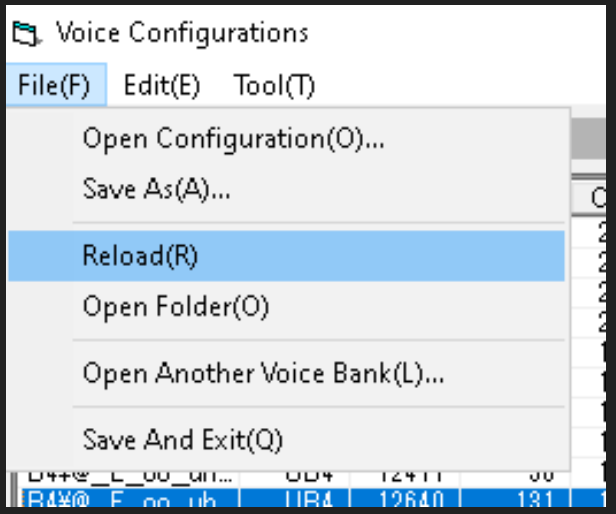
Creating OTOs

- There are several ways to make otoing GrayTalian much easier; the first of which is to use a base. For this reclist package, I have included a base configuration to use.
- It helps to generate FRQs for the samples before setting up the base oto, as there will be less FRQs to generate.
- SetParam is the program I use to generate configurations.
- Open SetParam and navigate to your bank's folder. When prompted to select configuration, choose "Load." Navigate to the oto.ini file provided in my reclist package. Load the configuration. **Note:** If the bank is multipitch, choose the oto.ini in the **"suffix"** subfolder. If monopitch, load regular config. After loading the base, navigate to File > Save Voice Configurations As… Make sure to save the configuration in the bank's correct folder, especially if it is multipitch.
- SetParam can be closed.









Creating OTOs (cont'd)

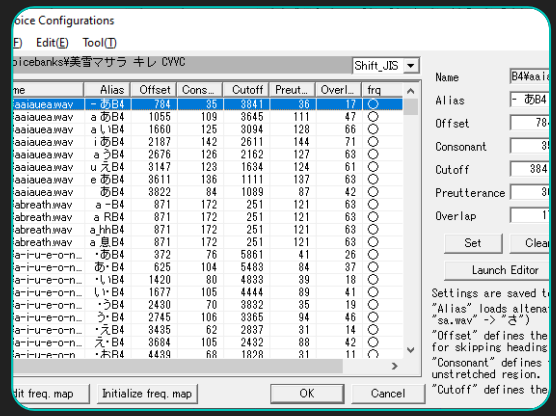
- If you are using the base oto for a multipitch bank, the suffix for all aliases can easily be changed at once.
- After generating all FRQs and applying the suffixed base oto in your voicebank's folder, open the oto.ini file in Notepad.
- Select Edit > Replace (CTRL + H)
- Change "Find what" to C#4, and "Replace with" to the correct pitch in the oto. Make sure "Match Case" is checked.
- Select "Replace all" to replace all suffixes.
- Select File > Save (CTRL + S) to save the oto.
- When opening voicebank configurations in UTAU again, make sure to select File > Reload to ensure the aliases change.

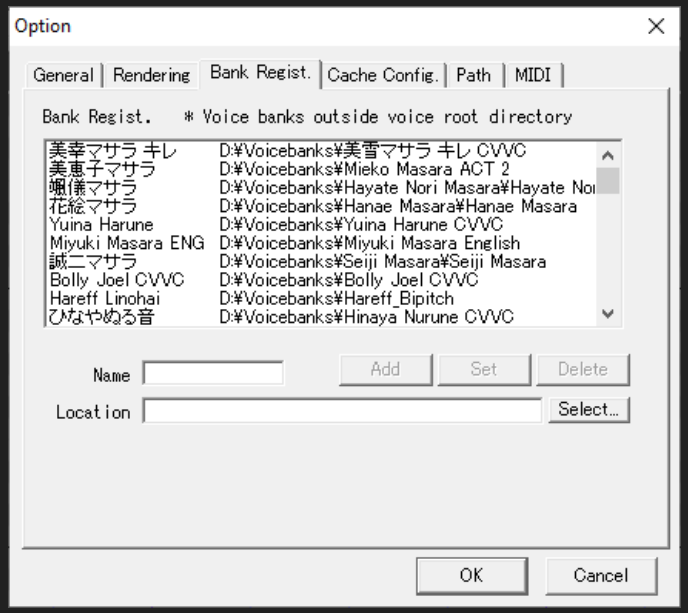
UTAU OTO

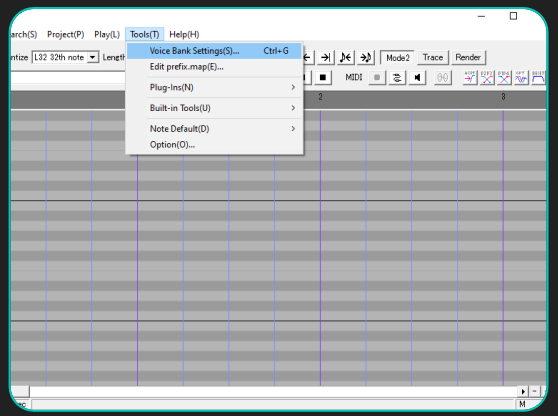
Open UTAU. Load your VB into UTAU by navigating to Tools > Option > Select (next to location) > Your bank's folder > Add > OK

Then navigate to Project (P) and select your UTAU from the drop down.

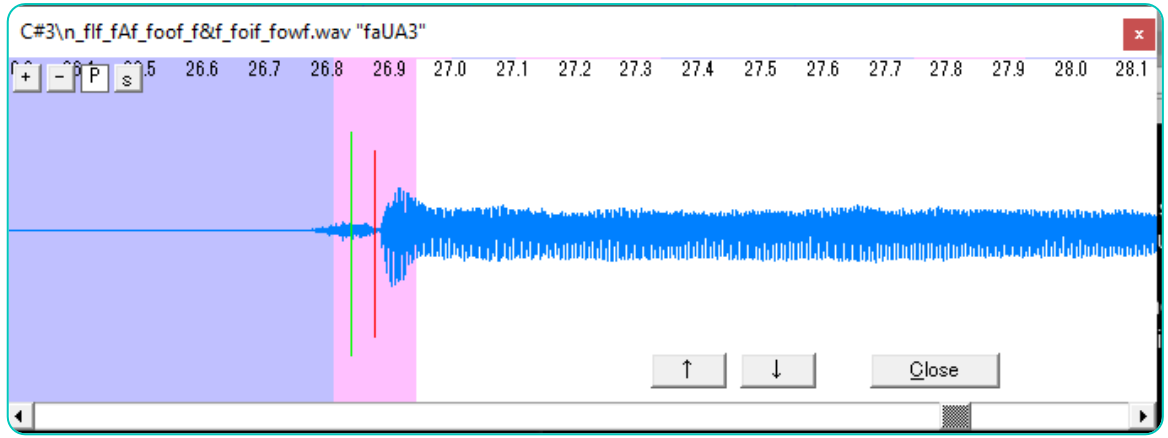
Open Tools > Voicebank settings.







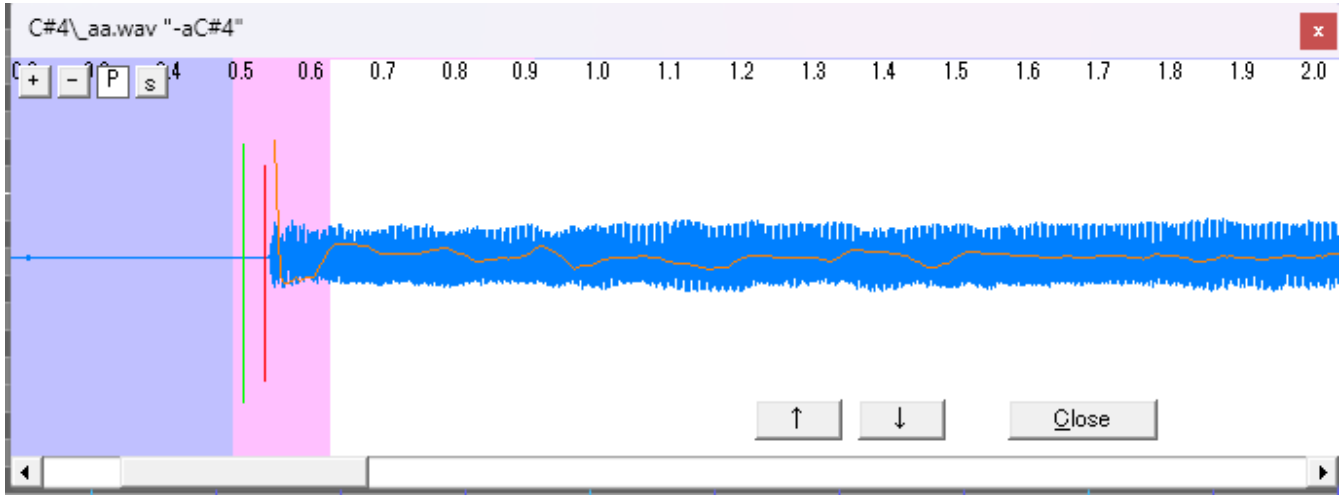
UTAU OTO (con'td)



- Pink part - "consonant"
- Red line: Preutterance
- Green line: Overlap
- Blue beginning part: Offset
- Blue ending part: Cutoff

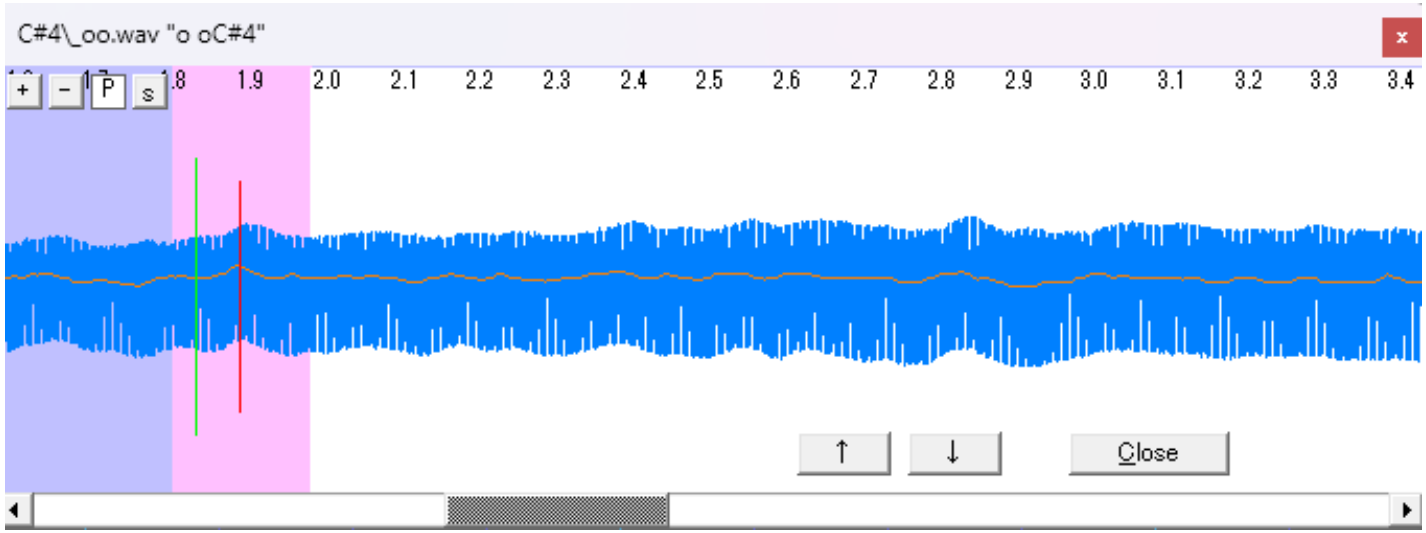
- The first blue part, offset, tells the audio where to begin
- Cutoff literally "cuts" the sample and tells it where to stop
- Preutterance is the sound that happens before the rest of the vowel or consonant (usage changes depending on what is being configured)
- Overlap is usually placed at half the length of the preutt.
- Consonant captures the initial part.

Standalone Vowels (-Vs)



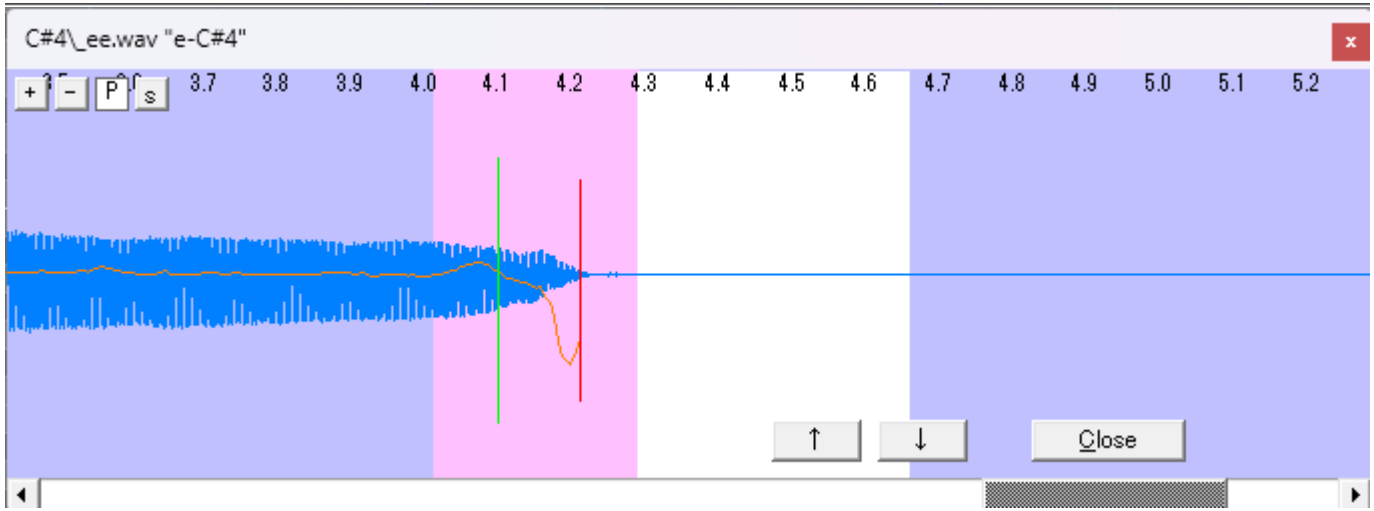
- For beginning vowels, **consonant** should be relatively small, to prevent the vowel from creeping too far into a rest note preceding it.
- **Offset** should allow for a little space between the period of silence and where the note begins, but not too much.
- **Preutterance** should **just** stop right before **consonant**.
- **Overlap** should stay about halfway between **offset** and **preutterance.**
- **Cutoff** should be placed right around the halfway point of the sample.

Standalone Vowels (V V)



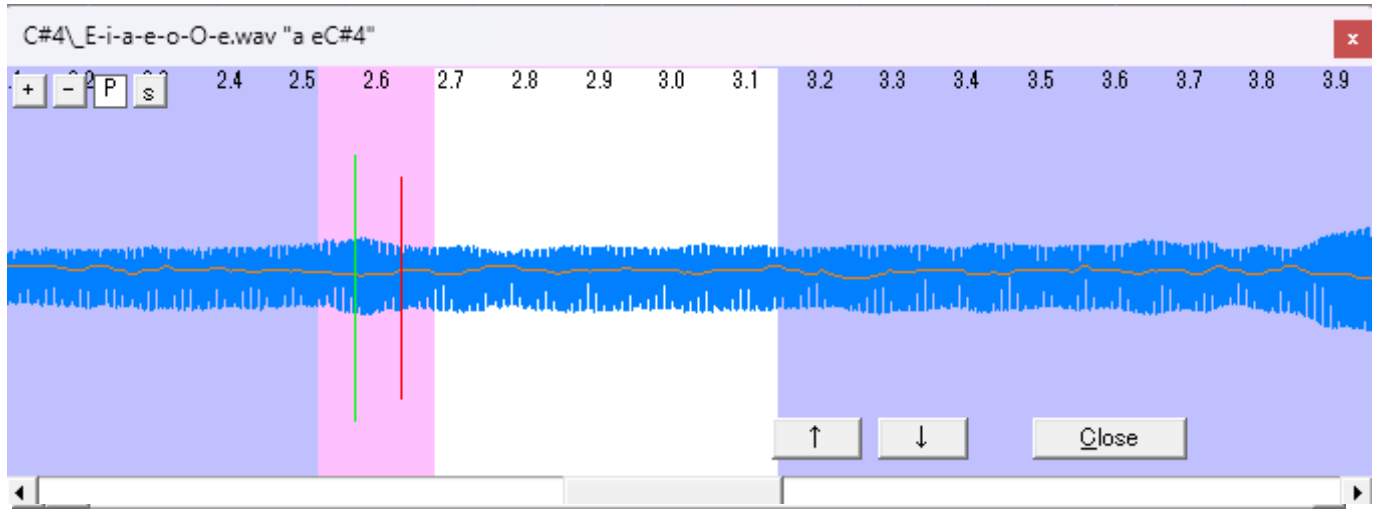
- For middle vowels (V Vs), **consonant** should still be somewhat small, but a bit longer than beginning ones.
- **Offset** should come in just a bit after a small portion of the first note and encompass the start of the second note.
- **Preutterance** should be **behind overlap** in the middle of **consonant**.
- **Overlap** should be just around where **consonant** ends.
- **Cutoff** should be placed before the note begins to taper off.

Standalone Vowels (V-)



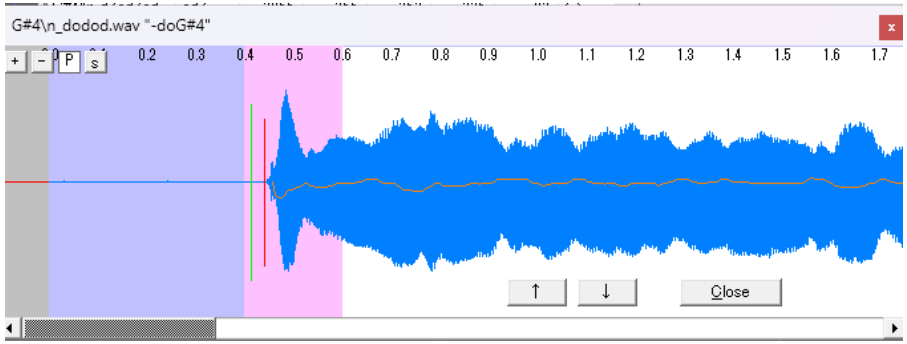
- For ending vowels, **consonant** should capture the ending bit of the vowel into a small portion of the breath.
- **Offset** should allow for small capture of the ending vowel.
- **Preutterance** should stop right after the vowel just before the breath begins.
- **Overlap** should stay about halfway between **offset** and **preutterance.**
- **Cutoff** should allow for a bit of silence after the end breath has ended.

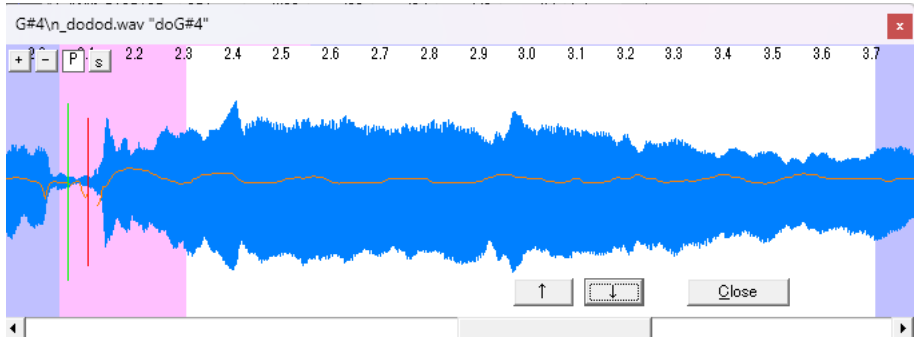
Vowel – Vowels (VVs)



- For vowel – vowel transitions, **consonant** should capture the small bit of the vowel before into a small portion of the second vowel.
- **Offset** should allow for a small capture of the first vowel.
- **Preutterance** should be after the first vowel and about halfway in between the **consonant.**
- **Overlap** should be right around where **consonant** ends**.**
- **Cutoff** should be placed where the second vowel ends and before the next vowel in the string begins.

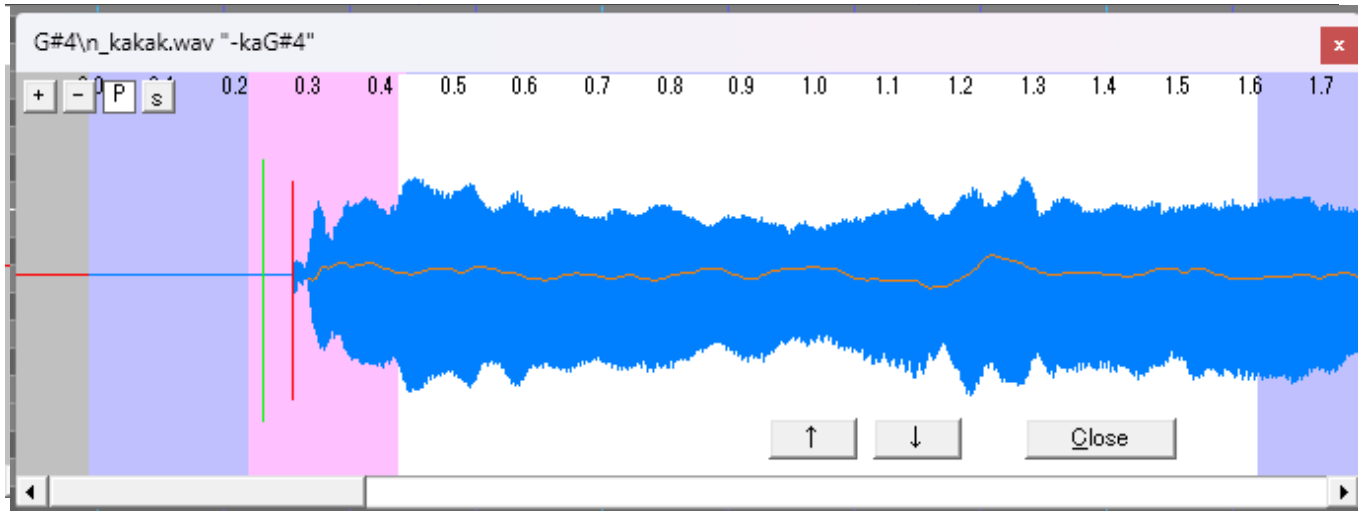
Main CVs (-CV, CVs)





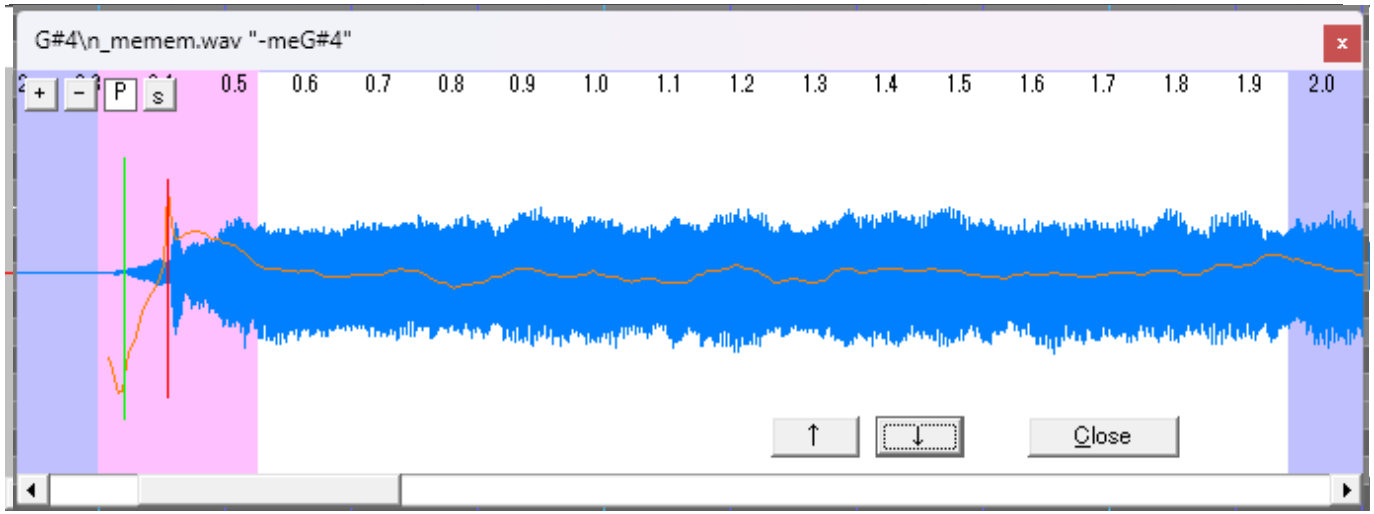
- For beginning consonant – vowel sounds with **consonant** should capture the consonant and a small portion of the following vowel.
- **Offset** should allow a small period of silence for consonant cutoff prevention, longer on –CVs but not too long.
- **Preutterance** should stop before the vowel.
- **Overlap** should stay about halfway between **offset** and **preutterance.**
- **Cutoff** should be placed before the second syllable on beginning -CVs and before the ending VC on the middle CVs.

Hard Consonants (CV)



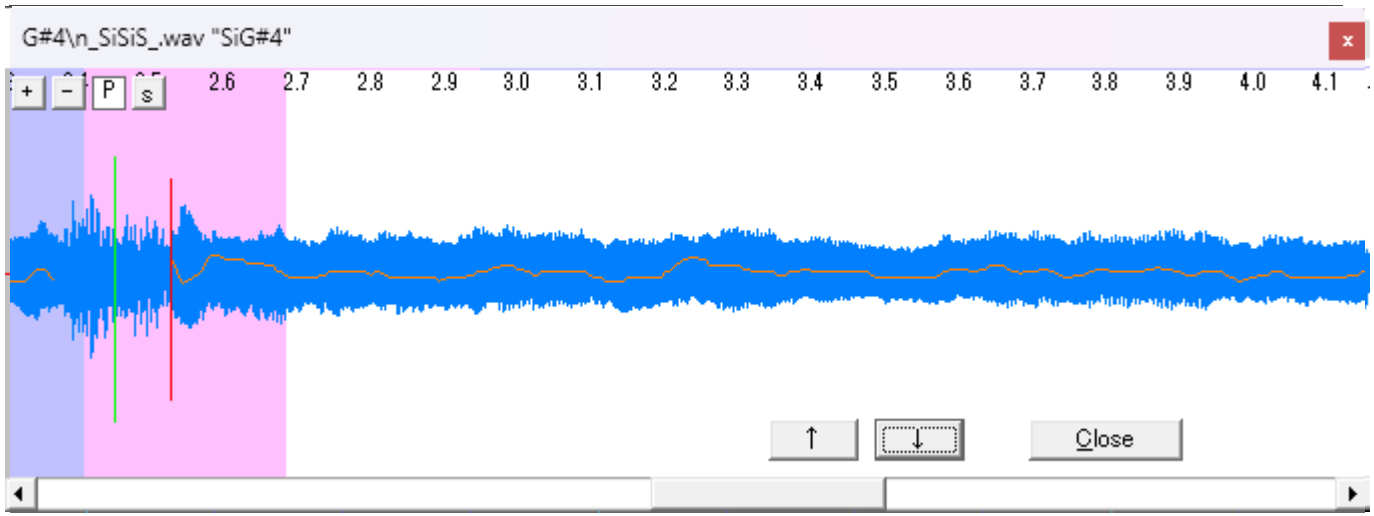
- For consonant – vowel sounds with **hard consonants**, **consonant** should capture the consonant and a small portion of the following vowel.
- **Offset** should allow a small period of silence for consonant cutoff prevention.
- **Preutterance** should stop before the vowel. If short, it can stop before the consonant on –CVs.
- **Overlap** should stay about halfway between **offset** and **preutterance.**
- **Cutoff** should be placed when the vowel ends and before the consonant begins.

Soft Consonants (CV)



- For consonant – vowel sounds with **soft consonants**, **consonant** should capture the consonant and a small portion of the following vowel.
- **Offset** should allow a small period of silence for consonant cutoff prevention.
- **Preutterance** should stop right after the consonant of the note and right in front of the vowel.
- **Overlap** should stay about halfway between **offset** and **preutterance.**
- **Cutoff** should be placed when the vowel ends and before the end consonant begins.

“S-Heavy” Consonants (CV)



- For consonant – vowel sounds with **“S-Heavy” consonants**, **consonant** should capture the consonant and a small portion of the following vowel.
- **Offset** may have to come into the consonant a little, as “S” consonants can become very long.
- **Preutterance** should stop right after the consonant of the note and right in front of the vowel.
- **Overlap** should stay about halfway between **offset** and **preutterance.**
- **Cutoff** should be placed when the vowel ends and before the end consonant begins.

Vowel – Consonants (VCs, V Cs, VCs-)

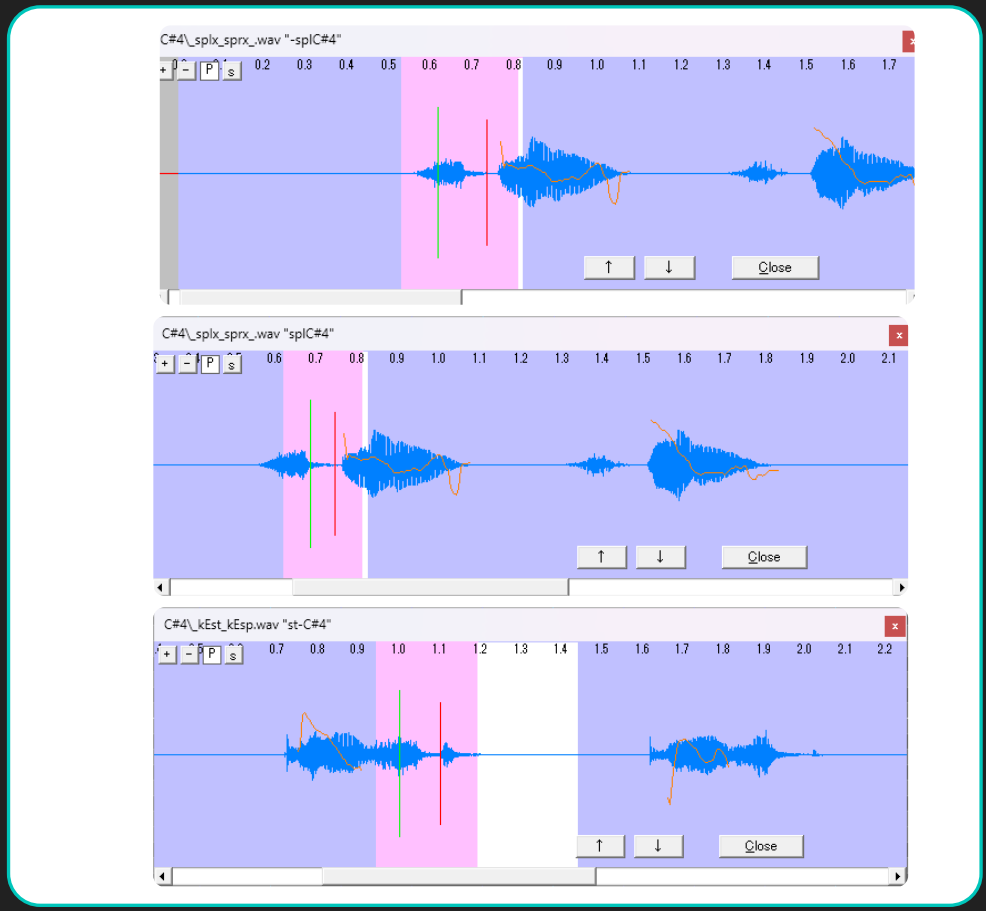
- For vowel – consonant sounds, **consonant** should capture vowel and stop before the ending consonant.
- **Offset** should allow for more vowel on V Cs, and less on VCs.
- **Preutterance** should stop right after the vowel of the note.

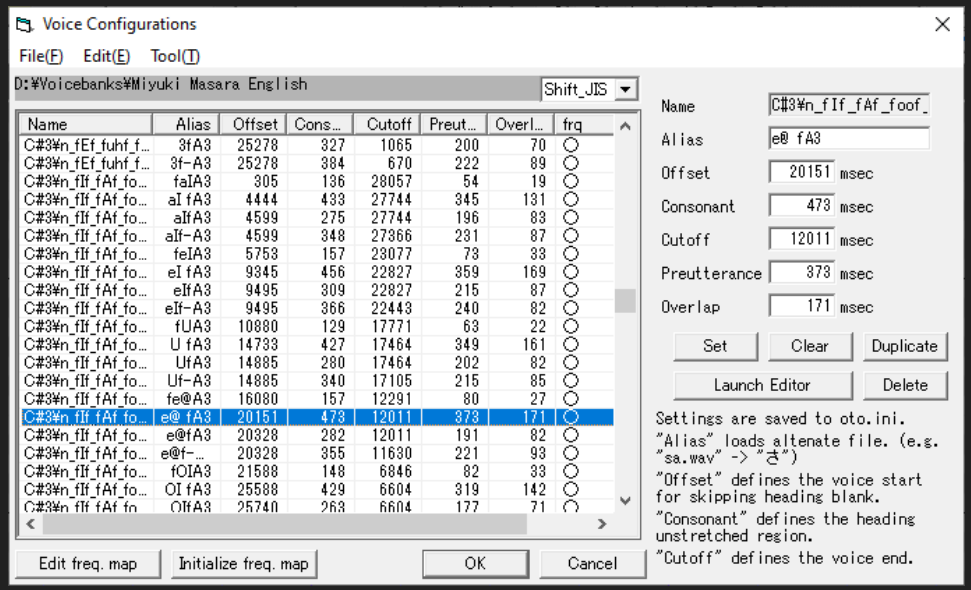
- **Overlap** should stay about halfway between **offset** and **preutterance.**
- **Cutoff** should be placed when the consonant ends on VCs and V Cs. For VCs-, **cutoff** should be allow for a period of silence after the ending consonant. A ratio of 1:1 is good to use for VCs-.

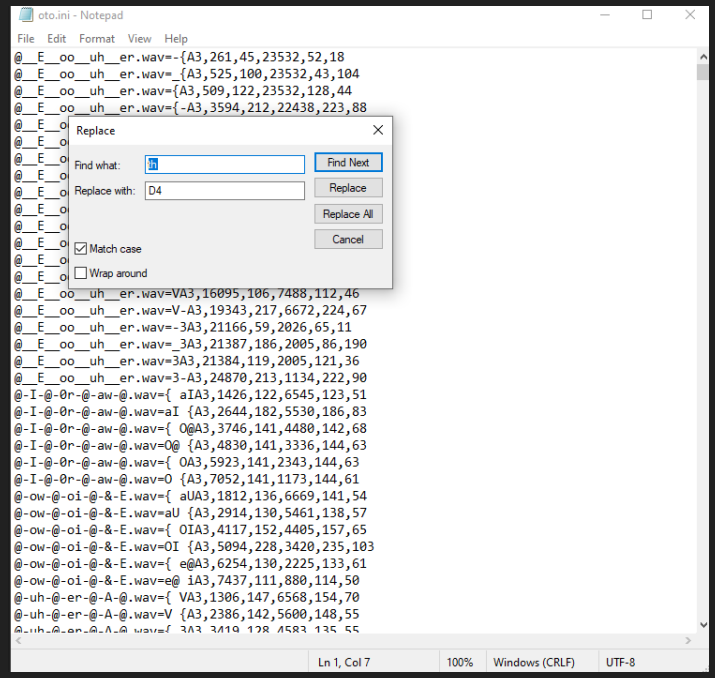


Consonant Clusters (-CC, CC, CC-s)

- For beginning consonant clusters, **consonant** should cover only up to the final consonant in the cluster.
- For middle consonant – consonant sounds, **consonant** should capture all consonant sounds before the final consonant in the note.
- For beginning CCs, **offset** should leave a small portion of silence before the note.
- In middle CCs, **offset** should include all consonant sounds, but not much of the vowel before.
- For beginning CCs, **preutterance** should stop after the first consonant in the cluster.
- For middle CCs, **preutterance** should stop right after the preceding consonants of the note and right before the final consonant.
- **Overlap** should stay about halfway between **offset** and **preutterance.**
- **Cutoff** should be placed when the final consonant ends in CCs. In CCs-, a period of silence with a 1:1 ratio should be used for **cutoff.**







Aliases

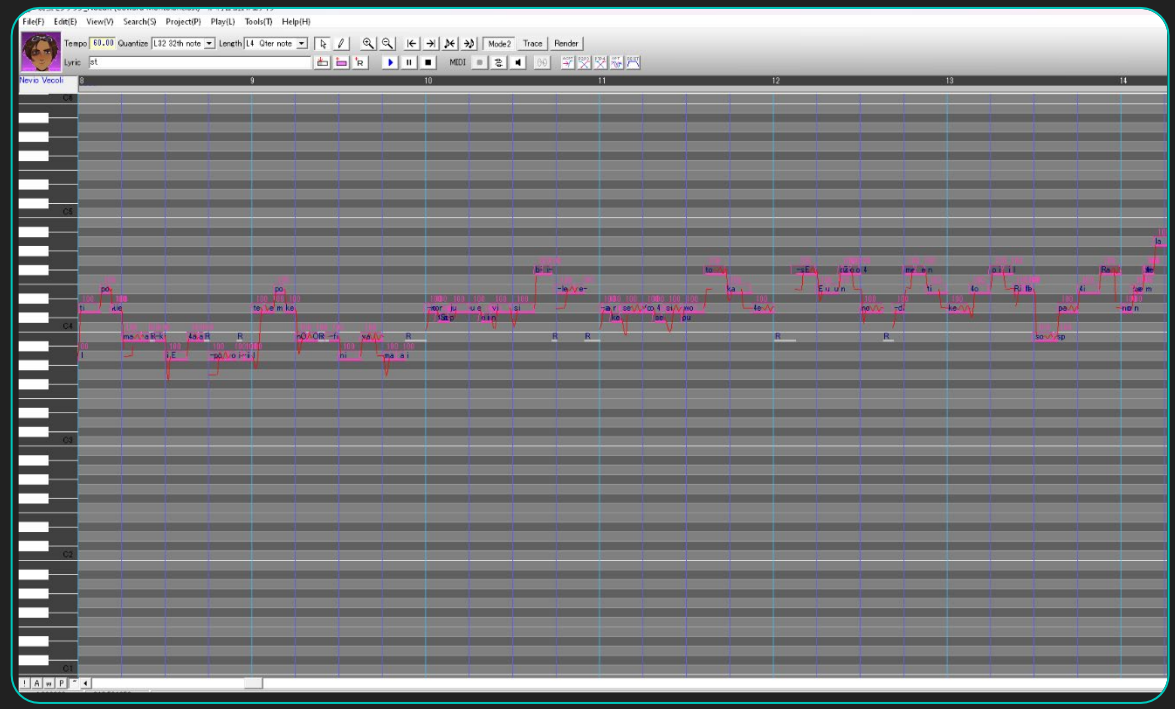
When otoing is complete or needs to be saved, press OK to close the voicebank editor.

Aliases are already set by the base oto, but may need to be edited if the bank is multipitch.

NOTE: The “Cancel” button does not save the oto or aliases.

Bank Testing

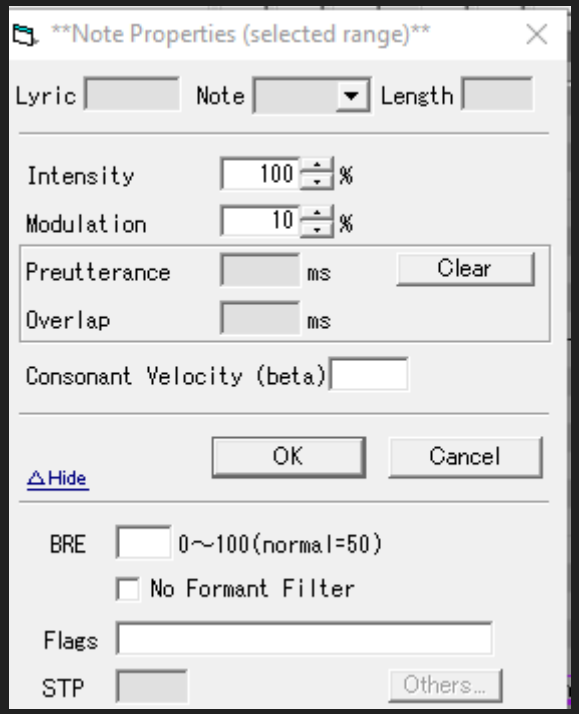
- We will be using the sample UST file that I have provided.
- Open the UST by navigating to File > Open > Open sample.UST.



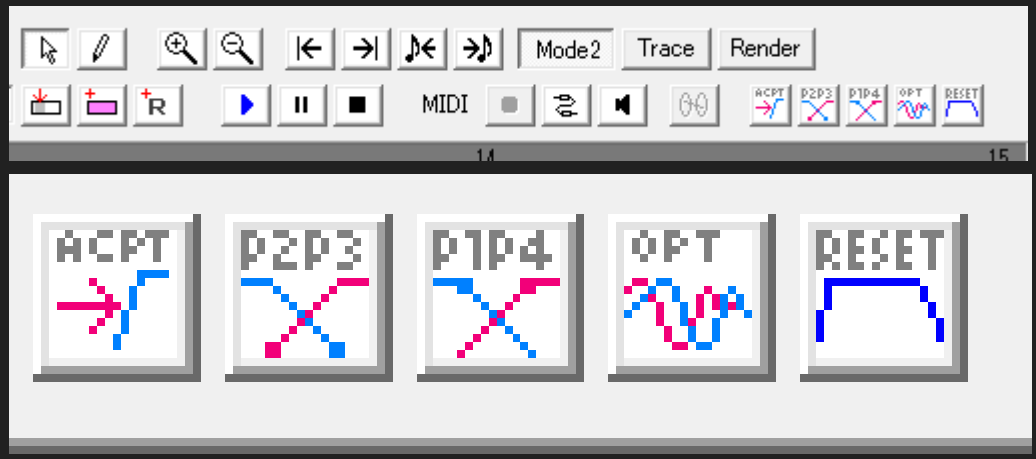
Fitting a UST

Make sure UTAU is in Mode 2!

<u>To finish fitting the UST:</u>



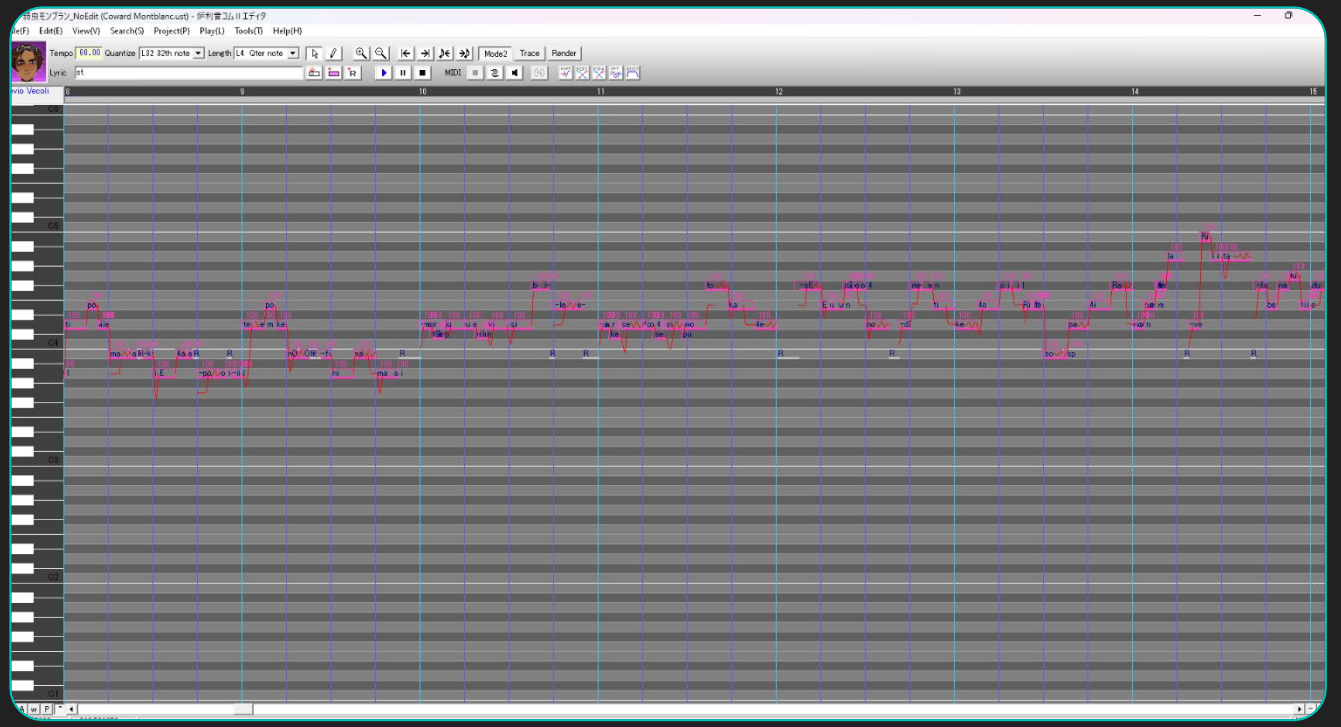
All USTs must be fitted to your UTAU's parameters. To do so, select all by pressing Ctrl+a. Press Ctrl+E to bring up the Region Property editor.



- Clear preutt. and overlap.
- If intensity is grayed out, set to 100 (default)
- Set modulation to 0.
- Leave consonant velocity blank for now.
- If BRE is grayed out, click twice to clear it out to the default 0.
- If flags box is grayed out, clear it.
- Clear STP (Starting point)
- Press OK when done.
- Click the fifth button to reset, then click the first button once, then the second, then the first again.

Playback

- **If all steps have been followed correctly**, you will have a flowing synthesized vocal.
- Congratulations! You now have a completed Grayglish bank.



Thank you!

- Thank you for reading this guide! I hope that it assisted you in making a GrayTalian voicebank.
- If you have any questions about this guide or reclist, I can be reached at the following:

